Trainings

Preparatory Steps

with Mechanically Actuated Injector

⚠ WARNING ⚠

Depending on the circumstance, fuel is flammable. Keep all cigarettes, flames, pilot lights, arcing equipment, and switches out of the work area and areas sharing ventilation to reduce the possibility of severe personal injury or death when working on the fuel system.

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

- Disconnect the battery cables. See equipment manufacturer service information.
- Remove the fuel filters.Refer to Procedure 006-015 in Section 6. (/qs3/pubsys2/xml/en/procedures/28/28-006-015-tr.html)

with Electronically Actuated Injector

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⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

- Disconnect the battery cables. See equipment manufacturer service information.
- Disconnect the fuel pressure sensor. Use the following procedure in the QSK38, QSK50 and QSK60 (CM850 Modular Common Rail System), Troubleshooting and Repair, Bulletin

4021533. Refer to Procedure 019-398 in Section 19.
(/qs3/pubsys2/xml/en/procedures/05/05-019-398.html)

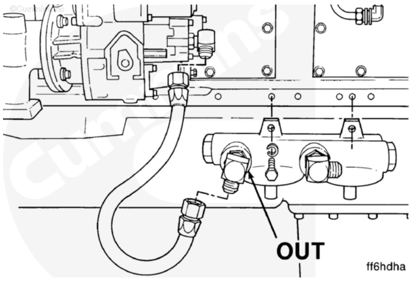
- Disconnect the fuel temperature sensor. Use the following procedure in the QSK38, QSK50 and QSK60 (CM850 Modular Common Rail System), Troubleshooting and Repair, Bulletin 4021533. Refer to Procedure 019-119 in Section 19. (/qs3/pubsys2/xml/en/procedures/05/05-019-119.html)
- Remove the fuel filters. Refer to Procedure 006-015 in Section 6. (/qs3/pubsys2/xml/en/procedures/28/28-006-015-tr.html)
- Disconnect the fuel supply lines. Refer to Procedure 006-024 in Section 6. (/qs3/pubsys2/xml/en/procedures/28/28-006-024-tr.html)

Remove

with Mechanically Actuated Injector

Remove the flexible hose.

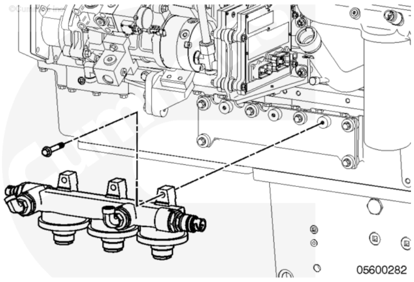
Remove the fuel filter head mounting capscrews and the fuel filter head.



with Electronically Actuated Injector

Stage 2 Filter Head

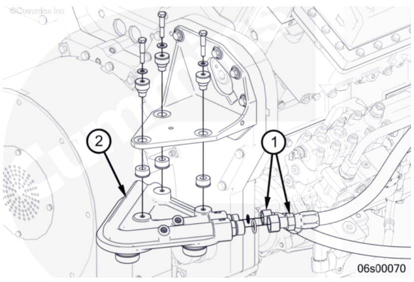
- Remove the fuel filter head mounting capscrews and the fuel filter head.



Front Mounted with Electronically Actuated Injector

Remove the two flexible hoses (1).

Remove the filter head mounting capscrews and rubber isolators along with the fuel filter head (2).



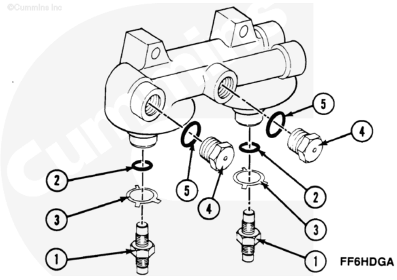
Disassemble

with Mechanically Actuated Injector

Remove the fuel filter adapters (1).

Remove the lockplates (3) and o-rings (2). Discard the o-rings.

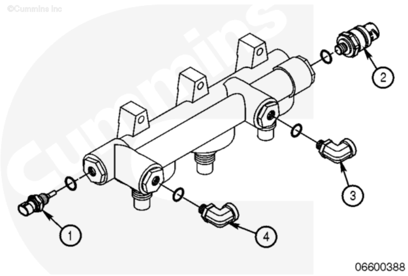
Remove the plugs (4) and o-rings (5). Discard the o-rings.



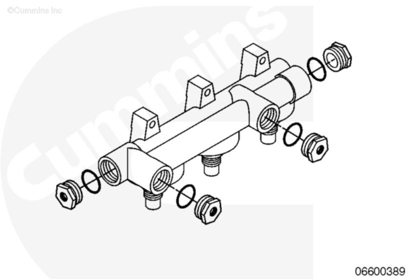
with Electronically Actuated Injector

Stage Two Filter Head

- Remove the fuel temperature sensor (1). Use the following procedure in the QSK38, QSK50 and QSK60 (CM850 Modular Common Rail System), Troubleshooting and Repair, Bulletin 4021533. Refer to Procedure 019-119 in Section 19. (/qs3/pubsys2/xml/en/procedures/05/05-019-119.html)
- Remove the fuel pressure sensor (2). Use the following procedure in the QSK38, QSK50 and QSK60 (CM850 Modular Common Rail System), Troubleshooting and Repair, Bulletin 4021533. Refer to Procedure 019-398 in Section 19. (/qs3/pubsys2/xml/en/procedures/05/05-019-398.html)
- Remove the fuel inlet (3) and fuel outlet (4) fittings.
- Discard the o-rings.

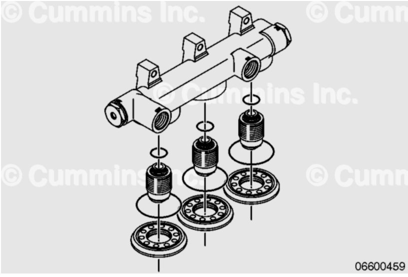


Remove the sensor and fitting adaptors. Discard all removed o-rings.



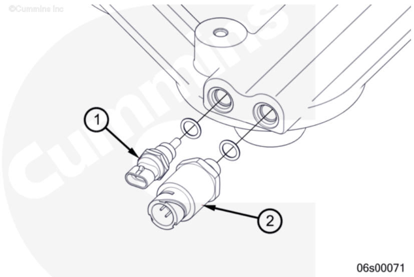
Remove the filter head adapter spuds and the adapter rings.

Remove and discard the six o-rings.



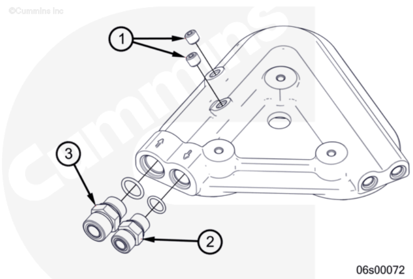
Front Mounted with Electronically Actuated Injector

- Remove the fuel temperature sensor (1). Use the following procedure in the QSK38, QSK50 and QSK60 (CM850 Modular Common Rail System), Troubleshooting and Repair, Bulletin 4021533. Refer to Procedure 019-119 in Section 19. (/qs3/pubsys2/xml/en/procedures/05/05-019-119.html)
- Remove the fuel pressure sensor (2). Use the following procedure in the QSK38, QSK50 and QSK60 (CM850 Modular Common Rail System), Troubleshooting and Repair, Bulletin 4021533. Refer to Procedure 019-398 in Section 19. (/qs3/pubsys2/xml/en/procedures/05/05-019-398.html)



- Remove the fuel inlet (3) and fuel outlet (2) fittings.
- Remove the two plug pipes (1)

Discard the o-rings.

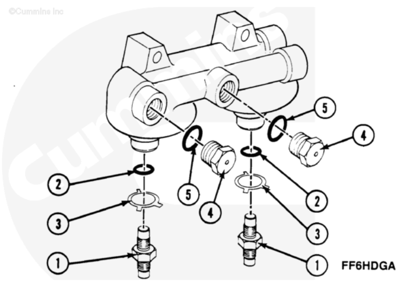


Clean and Inspect for Reuse

with Mechanically Actuated Injector

⚠ WARNING ⚠

When using solvents, acids or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

⚠ CAUTION ⚠

Use a solvent that will not damage aluminum.

Use solvent to clean the filter head components.

Use compressed air to dry.

Inspect the fuel filter adapters (1) for damaged or stripped threads. Replace if damage is found.

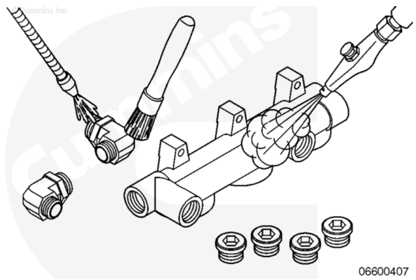
Inspect the plugs (4) for damaged or stripped threads. Replace if damage is found.

Inspect the lockplates (3) for missing tabs. Replace if damage is found.

with Electronically Actuated Injector

⚠ WARNING ⚠

When using solvents, acids or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

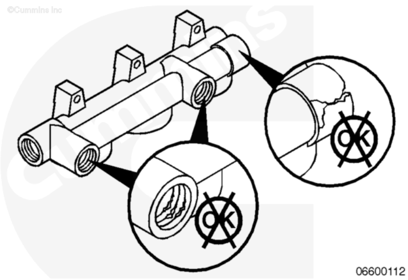
⚠ CAUTION ⚠

Use a solvent that will not damage aluminum or the remaining o-rings.

Stage Two Filter Head

- Use solvent to clean the filter head components.
- Use compressed air to dry.

Inspect the filter head for damaged threads or cracks in the housing.



Front Mounted with Electronically Actuated Injector

⚠ WARNING ⚠

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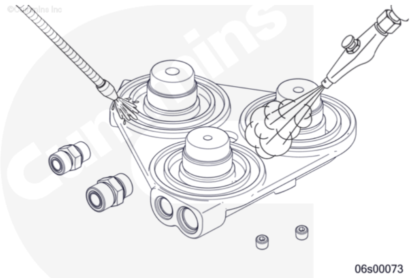
⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

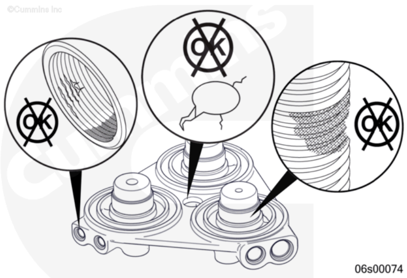
⚠ CAUTION ⚠

Use a solvent that will not damage aluminum or the remaining o-rings.

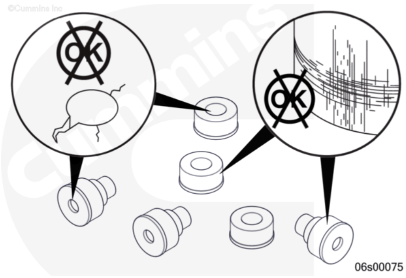
- Use solvent to clean the filter head components.
- Use compressed air to dry.



Inspect the filter head for damaged threads or cracks in the housing.



Inspect the filter head isolators for excessive wear or damage.

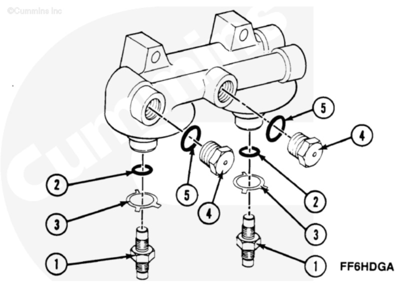


Inspect the fuel pressure sensor. Use the following procedure in the QSK38, QSK50 and QSK60 (CM850 Modular Common Rail System), Troubleshooting and Repair, Bulletin 4021533. Refer to Procedure 019-398 in Section 19. (/qs3/pubsys2/xml/en/procedures/05/05-019-398.html)

Inspect the fuel temperature sensor. Use the following procedure in the QSK38, QSK50 and QSK60 (CM850 Modular Common Rail System), Troubleshooting and Repair, Bulletin 4021533. Refer to Procedure 019-119 in Section 19. (/qs3/pubsys2/xml/en/procedures/05/05-019-119.html)

Assemble

with Mechanically Actuated Injector



Install new o-rings (2) and lockplates (3). Check that the lug that is bent at 90 degrees is seated in the drilled hole in the fuel filter head.

Install the lock plate. Locate the 90 degree bend in the drilled hole in the fuel filter head.

Apply one drop of Loctite® 601 on the threads of the fuel filter adapters (1) that engage the threads in the filter head, and install the adapter.

Torque Value: 102 n•m [75 ft-lb]

Note : The position of the plugs can be different than illustrated because of the various fuel plumbing arrangements.

Install the plugs (4) and new o-rings (5).

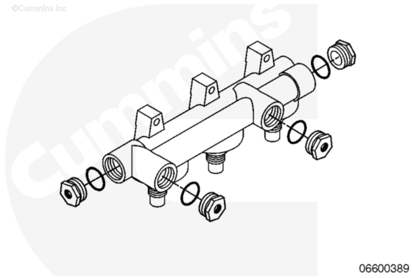
Torque Value: 102 n•m [75 ft-lb]

with Electronically Actuated Injector

Stage Two Filter Head

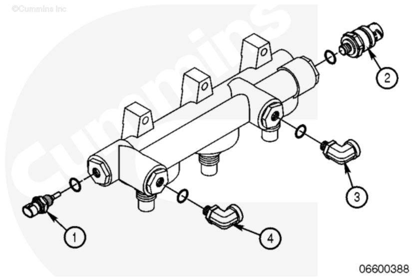
- Install the sensor and fitting adapters into the filter head.

Torque Value: 102 n•m [75 ft-lb]



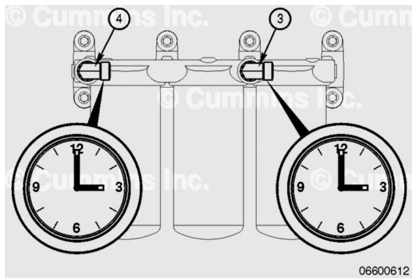
Install the fuel temperature sensor (1). Use the following procedure in the QSK38, QSK50 and QSK60 (CM850 Modular Common Rail System), Troubleshooting and Repair, Bulletin 4021533. Refer to Procedure 019-119 in Section 19. (/qs3/pubsys2/xml/en/procedures/05/05-019-119.html)

Install the fuel pressure sensor (2). Use the following procedure in the QSK38, QSK50 and QSK60 (CM850 Modular Common Rail System), Troubleshooting and Repair, Bulletin 4021533. Refer to Procedure 019-398 in Section 19. (/qs3/pubsys2/xml/en/procedures/05/05-019-398.html)



QSK50 Engines with Steel Braided Lines

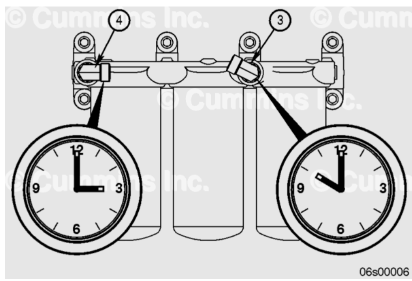
- Use new o-rings and install the fuel inlet (3) and fuel outlet (4) fittings.
- Make sure the elbow joints are positioned at a 3 o'clock angle, as shown.



Torque Value:
Elbow Fittings 115 n•m [85 ft-lb]

QSK50 Engines with Blue Nylon Lines

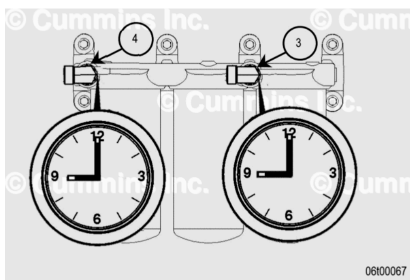
- Use new o-rings and install the fuel inlet (3) and fuel outlet (4) fittings.
- Make sure the elbow joints are positioned at the 3 o'clock and 10 o'clock angles, as shown.



Torque Value:
Elbow Fittings 115 n•m [85 ft-lb]

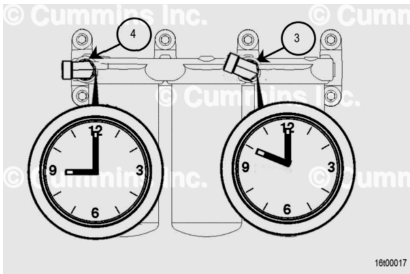
QSK38 Engine with Gerotor Fuel Pump Drain Line

- Use new o-rings and install the fuel inlet (3) and fuel outlet (4) fittings.
- Make sure the fuel outlet and inlet elbow joints are positioned at the 9 o'clock angles, as shown.



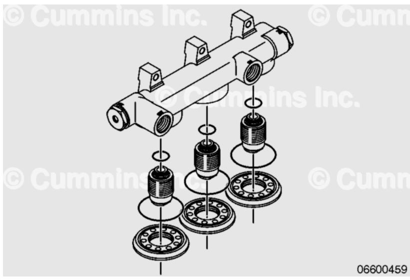
Torque Value:
Elbow Fittings 115 n•m [85 ft-lb]

QSK50 Engines with Gerotor Fuel Pump Drain Line



- Use new o-rings and install the fuel inlet (3) and fuel outlet (4) fittings.
- Make sure the elbow joints are positioned at the 9 o'clock (4) and 10 o'clock (3) angles, as shown.

Torque Value:
Elbow Fittings 115 n•m [85 ft-lb]



Install new o-rings above the filter head adapter spuds. Apply thread locker to the smaller threaded portion of the spud.

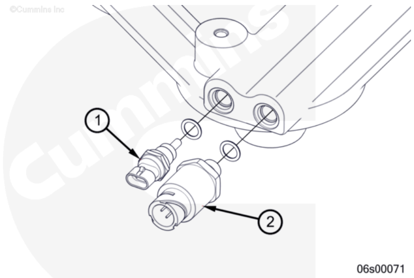
Install the spuds in the filter head.

Torque Value: 88 n•m [65 ft-lb]

Install new o-rings above the filter head adapter rings. Apply thread locking compound to the inside of the threads. Thread the rings onto the spuds until the o-rings contact the filter head. Use care to make sure the o-ring is properly seated.

Torque Value: 88 n•m [65 ft-lb]

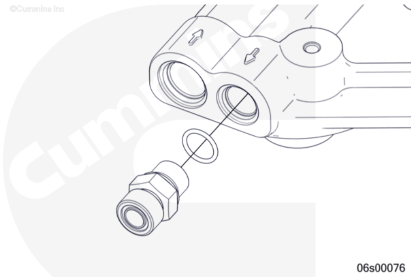
Front Mounted with Electronically Actuated Injector



- Install the fuel temperature sensor (1). Use the following procedure in the QSK38, QSK50 and QSK60 (CM850 Modular Common Rail System), Troubleshooting and Repair, Bulletin 4021533. Refer to Procedure 019-119 in Section 19. (/qs3/pubsys2/xml/en/procedures/05/05-019-119.html)
- Install the fuel pressure sensor (2). Use the following procedure in the QSK38, QSK50 and QSK60 (CM850 Modular Common Rail System), Troubleshooting and Repair, Bulletin 4021533. Refer to Procedure 019-398 in Section 19. (/qs3/pubsys2/xml/en/procedures/05/05-019-398.html)

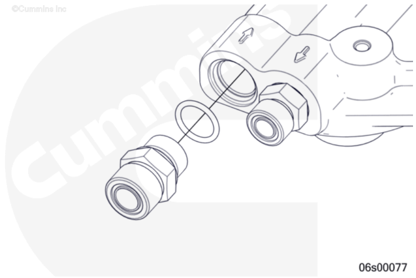
Using new o-rings, install the fuel outlet fitting.

Torque Value: 81 n•m [60 ft-lb]



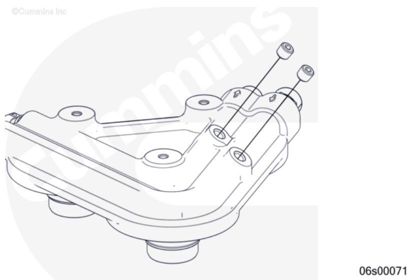
Using new o-rings, install the fuel inlet fitting.

Torque Value: 115 n•m [85 ft-lb]



Install the two pipe plugs. Apply Loctite® to the plug threads.

Torque Value: 25 n•m [221 in-lb]



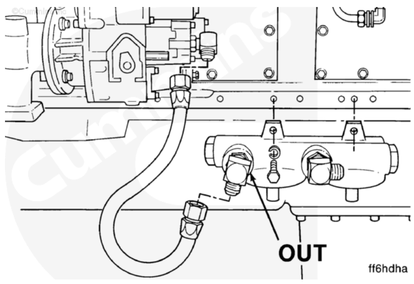
Install

with Mechanically Actuated Injector

Install the two lock washers and flat washers on the two capscrews.

Install the filter head and capscrews.

Torque Value: 55 n•m [41 ft-lb]



Note : The fuel filter head ports are marked “IN” and “OUT”.

Connect the hose for the port.

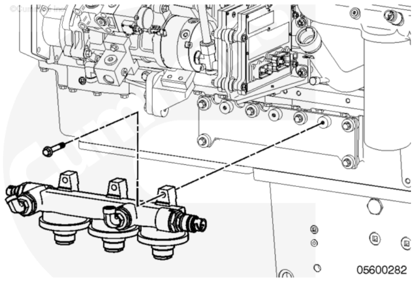
Connect the fuel pump inlet hose to the fitting on the filter head marked “OUT” and the fuel pump.

Tighten the hose fittings.

Torque Value:
Fuel Hose Fittings 55 n•m [41 ft-lb]

with Electronically Actuated Injector

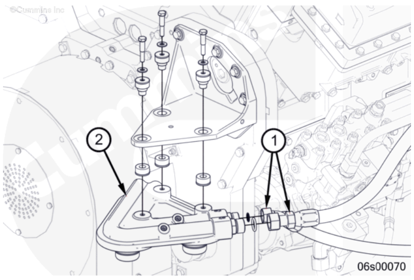
Stage 2 Filter Head



- Install the fuel filter head assembly to the engine with one nut and five mounting capscrews.

Torque Value: 45 n•m [33 ft-lb]

Front Mounted with Electronically Actuated Injector



Install the filter head assembly with isolators and capscrews (2).

Install the two flexible hoses (1).

Torque Value: 22 n•m [19 in-lb]

Finishing Steps

with Mechanically Actuated Injector

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

- Install the fuel filters. Refer to Procedure 006-015 in Section 6. (/qs3/pubsys2/xml/en/procedures/28/28-006-015-tr.html)
- Connect the battery cables. See equipment manufacturer service information.
- Operate the engine and check for leaks.

with Electronically Actuated Injector

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

- Install the fuel supply lines. Refer to Procedure 006-024 in Section 6. (/qs3/pubsys2/xml/en/procedures/28/28-006-024-tr.html)
- Install the fuel filters. Refer to Procedure 006-015 in Section 6. (/qs3/pubsys2/xml/en/procedures/28/28-006-015-tr.html)
- Connect the fuel temperature sensor. Use the following procedure in the QSK38, QSK50 and QSK60 (CM850 Modular Common Rail System), Troubleshooting and Repair, Bulletin 4021533. Refer to Procedure 019-119 in Section 19. (/qs3/pubsys2/xml/en/procedures/05/05-019-119.html)
- Connect the fuel pressure sensor. Use the following procedure in the QSK38, QSK50 and QSK60 (CM850 Modular Common Rail System), Troubleshooting and Repair, Bulletin 4021533. Refer to Procedure 019-398 in Section 19. (/qs3/pubsys2/xml/en/procedures/05/05-019-398.html)
- Connect the battery cables. See equipment manufacturer service information.
- Operate the engine and check for leaks.

Last Modified: 17-Mar-2020